

Lecture 21

From Equilibrium to Learning in Multi-Agent Systems

COMP532 – Machine Learning and Bioinspired Optimisation

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Today's Questions

At the end of this lecture, you should be able to answer:

1. Why is Nash equilibrium a useful but incomplete concept for multi-agent learning?
2. How can repeated interaction generate learning dynamics?
3. What is fictitious play, and what does it teach us about adaptation?
4. Why do learning processes sometimes converge, but sometimes cycle or fail?

Recap: What Lecture 20 Gave Us

In the previous lecture, we introduced:

- ▶ normal-form games
- ▶ best response
- ▶ dominant strategies
- ▶ Nash equilibrium
- ▶ stochastic / Markov games

What we now know

Game theory gives a language for **strategic stability**.

But one major question remains

How do agents actually **reach** an equilibrium through repeated interaction and learning?

Static Solution Concepts vs Dynamic Learning

Static viewpoint

- ▶ What outcomes are stable?
- ▶ What is a best response?
- ▶ What is a Nash equilibrium?

Dynamic viewpoint

- ▶ How do agents update?
- ▶ What do they learn from experience?
- ▶ Do their strategies converge?

Key distinction

An equilibrium describes a **state**. Learning describes a **process**.

Why Equilibrium is Not the End of the Story

Even if a Nash equilibrium exists, this does not imply that agents will find it.

Important open questions include:

- ▶ What information do agents have about others?
- ▶ Do they know the full game?
- ▶ Do they best-respond exactly or only approximately?
- ▶ Do updates happen synchronously or asynchronously?
- ▶ If multiple equilibria exist, which one is selected?

Main message

Equilibrium analysis tells us what *could* be stable. Learning analysis asks what agents *actually do over time*.

Repeated Interaction Changes the Problem

A one-shot game captures a single strategic interaction.

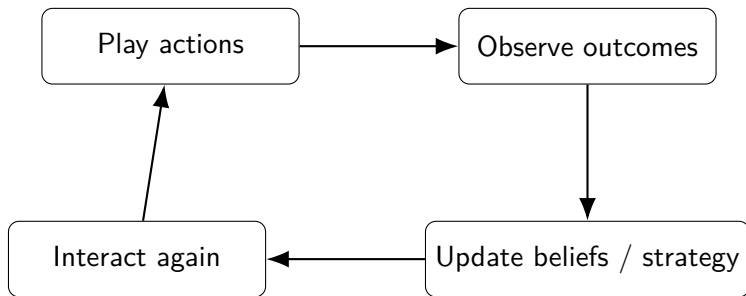
A repeated game allows agents to:

- ▶ observe past actions
- ▶ update beliefs
- ▶ adapt strategies over time
- ▶ react to patterns in the behaviour of others

Implication

Repeated interaction creates the possibility of **learning dynamics**.

A Generic Learning Loop in Games



Core idea

Learning in games is an iterative feedback process between **interaction**, **observation**, and **adaptation**.

Beliefs About Other Players

To act strategically, an agent often needs beliefs about how others behave.

For example, an agent may estimate:

- ▶ how often another agent chooses each action
- ▶ whether their behaviour is stationary or changing
- ▶ whether they are rational, noisy, or exploratory

Central difficulty

In multi-agent learning, the target of learning is often another learner.

Fictitious Play: Basic Idea

Fictitious play is a classical model of learning in games.

At each round:

1. each player observes the empirical frequencies of others' past actions
2. each player assumes the others will continue to play according to those frequencies
3. each player chooses a best response to that empirical belief

Interpretation

Players behave as if the opponents were stationary distributions estimated from past play.

Fictitious Play as Belief-Based Learning

Let $\hat{\pi}_{-i}^{(t)}$ denote player i 's empirical estimate of the others' strategy at time t .

Then player i chooses:

$$a_i^{(t)} \in \arg \max_{a_i \in \mathcal{A}_i} u_i(a_i, \hat{\pi}_{-i}^{(t)}).$$

After observing the next action, the empirical estimate is updated.

Key insight

Fictitious play turns repeated interaction into a sequence of **belief updates + best responses**.

Why Fictitious Play is Important

Fictitious play is not just one specific algorithm. It is a conceptual bridge between:

- ▶ static game theory
- ▶ repeated interaction
- ▶ adaptive learning behaviour

It helps us ask:

- ▶ If players repeatedly best-respond to experience, what happens?
- ▶ Will the system settle down?
- ▶ Can learning itself be analysed as a dynamical process?

Example: Coordination Game

	A	B
A	(3,3)	(0,1)
B	(1,0)	(2,2)

If both players observe that the other has recently chosen **A** often, then:

- ▶ best response is to play **A**
- ▶ this reinforces further play of **A**

Similarly, if play drifts toward **B**, that can also become self-reinforcing.

Observation

Learning dynamics can amplify early patterns and lead to equilibrium selection.

Example: A Game That May Cycle

Consider a competitive setting such as matching pennies or rock-paper-scissors.

In such games:

- ▶ a best response to the current opponent action creates pressure for the opponent to change
- ▶ once the opponent changes, the previous best response may no longer be good
- ▶ this can produce oscillation rather than convergence

Important lesson

Best-response adaptation does not always settle to a fixed point in pure strategies.

Convergence vs Cycling

Learning dynamics in games can display several qualitatively different behaviours:

1. **Convergence** to a stable equilibrium
2. **Coordination** on one of several equilibria
3. **Oscillation** or cycling
4. **Instability** under changing beliefs or updates

Main point

To understand multi-agent learning, we must study not only *what equilibria exist*, but also *what the update dynamics do*.

Why Learning in Games is Hard

Several issues complicate learning dynamics:

- ▶ agents may observe only actions, not payoffs or policies
- ▶ agents may update simultaneously
- ▶ observations may be noisy
- ▶ opponents may also be learning and exploring
- ▶ multiple equilibria may create path dependence

Key challenge

From each agent's viewpoint, the environment is strategic and potentially non-stationary.

What Fictitious Play Assumes

Fictitious play is elegant, but its assumptions are strong.

Typical assumptions include:

- ▶ repeated interaction in the same game
- ▶ the ability to observe others' actions
- ▶ the ability to compute a best response
- ▶ the idea that empirical frequencies are meaningful summaries

Why this matters

These assumptions become harder to satisfy in large-scale or sequential MARL settings.

From Fictitious Play to Modern Multi-Agent Learning

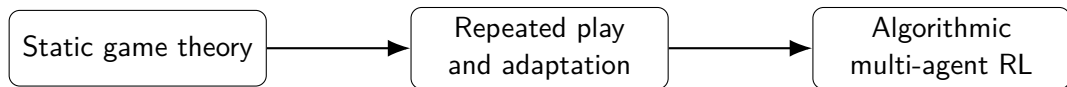
Fictitious play already contains several ideas that reappear in modern MARL:

- ▶ opponent modelling
- ▶ belief updating
- ▶ iterative adaptation
- ▶ equilibrium-seeking behaviour

But modern MARL must additionally deal with:

- ▶ large state spaces
- ▶ sequential decisions
- ▶ function approximation
- ▶ partial observability

A Conceptual Spectrum



- ▶ Static game theory asks what stable outcomes look like
- ▶ Repeated play asks how behaviour changes over time
- ▶ MARL asks how agents can learn in complex environments

Summary

1. Nash equilibrium is a static concept of strategic stability.
2. Repeated interaction creates learning dynamics.
3. Fictitious play is a classical model of belief-based adaptation.
4. Learning in games may converge, coordinate, oscillate, or fail.
5. Understanding dynamics is essential before studying MARL algorithms.

Looking Ahead

Next lecture:

- ▶ How do modern multi-agent RL algorithms work?
- ▶ What happens if each agent simply runs RL independently?
- ▶ When do we need explicit reasoning about the joint action or equilibrium?

Bridge to Lecture 22

We now move from **conceptual learning dynamics** to **algorithmic multi-agent reinforcement learning**.